MILITARY SPECIFICATION

MICROFILMING OF ENGINEERING AND RELATED DOCUMENTS, 16MM REQUIREMENTS FOR

1. SCOPE

1.1 Scope. This specification covers the detail requirements for microfilming of engineering and related documents, and the preparation of reproduction microfilm by the Department of Defense, or by contractors. Microfilm produced in accordance with requirements herein shall be 16mm unperforated roll microfilm.

1.2 Classification.

1.2.1 16mm Microfilm. 16mm microfilm shall be classified according to the film type, generation, and image appearance (Type, Class and Subclass, respectively).

| | | |
|---|---|---|
| Type | I | Silver-halide microfilm, Federal Specification L-F-315. |
| Type | II | Diazotype microfilm, Federal Specification L-F-315. |
| Type | III | Thermal developing microfilm (vesicular) Federal Specification L-F-320. |
| Class | 1 | Original camera master microfilm (Type I only) first generation. |
| Class | 2 | Distribution copy film (Types I, II or III), from Class 1, second generation. |
| Class | 3 | Distribution copy film (Types I, II or III), from Class 2, third generation. |
| Class | 4 | Distribution copy film (Types I, II or III), from Class 3, fourth generation. |

Area: EDMS

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Style N Negative-appearing image. Clear line image (Numerals, characters, etc.) on a dark background.

Style P Positive-appearing image. Dark line image on a light background.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect, on the date of invitation for bid or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

L-F-315 - Film Diazotype Sensitized (Direct Image Microfilm)

L-F-320 - Film, Thermal Developing

L-F-334 - Film, Photographic, Roll Microfilm (Black and White)

PPP-B-636 - Box, Fiberboard

P-W-120 - Wax, Automobile Paste

Military

MIL-M-9868 - Microfilming of Engineering Documents, 35MM Requirements for

MIL-M-38761 - Microfilm and Photographing of Engineering Technical Data and Related Documents: PCAM Card Preparation, Engineering Data Microreproduction System, General Requirements for Preparation of

STANDARDS

Federal

FED-STD-125 - Film, Photographic and Film, Photographic Processed; (For Permanent Record Use)



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FED-STD-595 - Color (Requirements for individual color chips 3X5 Supplements) should be submitted to the Navy Supply Depot, Philadelphia, Pa., citing the color chip number as specified in this specification. This specification requires color chip number 37038 (Black) and 37875 (White).

Military

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and forms required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other Publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

2.2.1 American National Standards Institute, Inc. (ANSI). Standards may be obtained from the American National Standards Institute, 1430 Broadway, New York, N.Y. 10018

ANSI PH2.19-1959 - Diffuse Transmission Density

ANSI PH4.8 -1971 - Determining Thiosulfate Content of Processed Black and White Photographic Film and Plates, Method for

ANSI PH5.6 -1968 - 100 Foot Reels for Processed 16mm and 35mm Microfilm Dimensions for

(Technical societies and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)



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2.2.2 National Bureau of Standards (NBS). The Microcopy Resolution Test Chart may be obtained from the Standard Materials Unit, National Bureau of Standards, Washington, D. C. 20234.

Microcopy Resolution Test Chart, Standard Sample 1010a

3. REQUIREMENTS

3.1 General. The general requirements for processed 16MM microfilm are contained in Section 3, of MIL-M-38761. The processed microfilm shall be furnished on reels or in containers as specified (See 6.2).

3.2 Classification of Requirements.

Requirements

Microfilm.....3.3
Documents.....3.4
Copy Board.....3.5
Container.....3.6
Reel.....3.7
Targets.....3.8
Supporting Documentation.....3.9
Security.....3.10

3.3 Microfilm.

3.3.1 Processing. Unless otherwise specified (See 6.2), all silver-halide microfilm shall be processed in accordance with FED-STD-125, and tested in accordance with 4.5.

3.3.2 Workmanship. Microfilm produced in accordance with this specification shall be free of scratches, holes in the emulsion or base, finger marks, or any other defect that might adversely affect the quality of reproductions made from microfilm.

3.3.3 Storage. All unexposed microfilm shall be stored in a cool dry area in accordance with the manufacturers' recommendations until used. Where microfilm is stored in refrigeration, it shall be allowed to adjust to room temperature for twelve hours before use.



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3.3.4 Leader. The leader mounts on the front of the film by splicing in accordance with 3.3.6. Film leaders shall not be interchanged between containers made by different manufacturers. The proper length of the leader is furnished by the manufacturer.

3.3.5 Trailer. The trailer mounts to the container core or reel and end of film. The same caution depicted for the leader as to interchangeability and splicing applies to the trailer (See 3.3.4 and 3.3.6). The proper length of the trailer is furnished by the manufacturer.

3.3.6 Splicing. The manufacturer of the container and the applicable loading station furnishes complete instructions on the proper method for splicing of film which meets the engineering design requirements for his containers and readers, and all splicing shall be performed according to such instructions. All splicing shall be checked by the contractor to assure that the splice adheres to the film; that no detrimental air bubbles or foreign particles are under the splice, and the splice does not make illegible any part of the code, document mark, timing mark, on the film. The splice shall have a maximum density of .10 and the opaque splice shall have a minimum density of 1.50. Unless otherwise specified (See 6.2), splicing within other parts of the roll shall not be permitted. (See figure 1.)

3.3.7 Frame Numbering. The data on the microfilm roll shall be arranged as follows:

a. Security Classification, if any (See 3.8.3)
b. Resolution Target (Fig 2)
c. Density Target (Fig 9)
d. Index, if any (See 3.3.14)
e. Documents

A frame number shall be placed on every frame of microfilm. The frames shall be numbered consecutively beginning with the frame bearing the security classificaion, if any; otherwise, the resolution target frame shall be numbered first and all other frames i.e., density target, index, and documents, shall be filmed consecutively. If there is no index, the first document number shall follow that of the density target frame. The numbering of every frame may not be required if specified in 6.2.



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3.3.7.1 Frame Number Area Size and Location. Unless otherwise specified (see 6.2), the frame number area size and location shall be within the dimensions specified in Figure 4.

3.3.8 Resolution. The microfilm resolution shall be established by microfilming the resolution target (Figure 2) and the density target (Figure 9) at a 24:1 reduction ratio (2 frames total). After a density between 1.0 and 1.2 has been obtained from the 50% reflectance chart in Figure 9, the National Bureau of Standards, Microcopy Resolution Test Chart, Standard Sample 1010a located in Figure 2 shall as a minimum resolve:

Original (first generation) camera microfilm (1N or 1P).....5.0
Second generation (2N or 2P).....4.5
Third generation (3N or 3P).....4.0
Fourth generation (4N or 4P).....3.6

3.3.9 Readability. In order for data to be readable from the microfilm and the microfilm reader the following must be considered:

Original document type point size
Reduction ratio of the data to microfilm
Lens reduction ratio between camera and reader
Optical quality of comera and reader lens

The smallest type point size that can be microfilmed and remain readable for this specification shall be as specified in 3.3.9.1 and 3.3.9.2.

3.3.9.1 16MM Microfilm Reduction Ratio and Type Point Size. Unless otherwise specified (see 6.2), the reduction ratios for 16MM microfilm shall be in accordance with Figure 10. The type point size shall be:

| Camera Reduction Ratio | Type Point Size (Figure 3 for height dimension) |
|---|---|
| 8:1 to 19:1 | 4 |
| 20:1 to 24:1 | 5 |

3.3.9.2 35MM Microfilm Reduction Ratio and Type Point Size. Unless otherwise specified (see 6.2), the reduction ratios for 35MM microfilm shall be as specified in Figure 19. The type point size shall be:

| | |
|---|---|
| 8:1 | 6 |
| 9:1 to 10:1 | 7 |
| 11:1 | 8 |
| 13:1 to 14:1 | 9 |
| 15:1 to 16:1 | 10 |
| 17: to 19:1 | 11 |

} Type Point Size (Fig 3 for height dimension)



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3.3.10 Type Point Sizes. Since there are hundreds of type fonts which vary between manufacturers, any type of font that meets the dimensional requirements of Figure 3 is acceptable.

3.3.11 Retrieval Codes. Unless otherwise specified (See 6.2) all microfilm shall be coded or uncoded as follows:

3.3.11.1 Uncoded Code Style S1. Uncoded microfilm code style S1 shall meet the dimensions specified in Figure 4, exclusive of the document mark.

3.3.11.2 Uncoded (Short Length) Code Style S2. Uncoded (Short Length) microfilm code style S2 shall meet the dimensions specified in Figure 5.

3.3.11.3 Bar Coded, Code Style S3. Bar coded microfilm code style S3 shall meet the dimensions specified in Figure 8.

3.3.11.4 Document Count, Code Style S4. Document count (frame count) microfilm code style S4 shall meet the dimensions specified in Fig. 4.

3.3.11.5 Document Count, (Short Length) Code Style S5. Document count (short length) microfilm code style S5 shall meet the dimensions specified in Fig 7.

3.3.11.6 Binary-Excess-Three, Code Style S6. Binary-excess-three microfilm code style S6 shall meet the dimensions specified in Figure 6.

3.3.11.7 Bar Coded (Short Length) Code Style S7. Bar coded (short length) microfilm code style S7 shall meet the dimensions specified in Fig. 13.

3.3.11.8 Binary-Excess-Three (Short Length) Code Style S8. Binary-excess-three microfilm code style S8 shall meet dimensions specified in Figure 14.

3.3.11.9 Non-Standard Microfilm Codes. The non-standard microfilm codes are assigned as follows:

3.3.11.9.1 Bar Code Between Microfilm Frames Code Style S9. This style of microfilm shall be as specified (See 6.2).

3.3.11.9.2 Microfilm Frame Length and Width As Specified Code Style S10. This style of microfilm shall be as specified (See 6.2)

3.3.12 Sequence. Unless otherwise specified (See 6.2) all documents shall be in ascending page number sequence.

3.3.13 Density. All densities specified in this specification are gross densities. The density shall be visual diffuse density (Type VI B) as measured using the method described in ANSI PH2.19-1959.

3.3.13.1 Background. The background density of processed silver-halide microfilm shall measure:



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| | DENSITY |
|---|---|
| Camera Negative Image Master Microfilm (first generation) | 1.0 - 1.2 |
| Negative Image Copy (second generation) | 0.90 - 1.30 |
| Negative Image Copy (third generation) | 0.90 - 1.30 |
| Negative Image Copy (fourth generation) | 0.85 - 1.35 |

| | DENSITY (Maximum) |
|---|---|
| Camera Positive Image Master Microfilm (first generation) | 0.16 |
| Positive Image Copy (second generation) | 0.20 |
| Positive Image Copy (third generation) | 0.20 |
| Positive Image Copy (fourth generation) | 0.24 |

3.3.13.2 Unexposed Clear Film Density. Processed clear film shall have a maximum density reading of 0.16. To determine such a density, an image reversing film shall be unexposed and processed and a direct image film shall be completely exposed and processed.

3.3.13.3 Opaque Code. Opaque code shall have a density of not less than 1.15 above the base density of the surrounding area. The opaque density shall not be allowed to bleed (overlap) into the adjacent clear area.

3.3.14 Index. When the procuring activity includes the requirement for a microfilm index (See 6.2), the index shall be microfilmed as specified on 3.3.7.

3.3.15 Area. The microfilm areas shall be as follows:

3.3.15.1 Frame. The microfilm frame area for coded microfilm shall be dimensions "A" (width) and "B" (length) as specified on Figures 4, 6 and 8. Dimensions "A-2" (width) and "B-2" (length) are specified on Figure 7. For uncoded microfilm, the dimensions "A" (width) and "B" (length) are specified on Figure 4 (excluding the document mark). Dimensions "A-1" (width) and "B-1" (length) are specified on Figure 5.

3.3.15.2 Document. The microfilm document area shall be determined as specified in 3.4.4 and Figure 10.

3.3.15.3 Frame Number. Unless otherwise specified (See 6.2), the frame number area shall be dimension "K" (width) and "G" (length) as specified on Figures 2, 4, 5, 6, 7, 8 and 9.

3.3.15.4 Document Mark. Unless otherwise specified (See 6.2), the document mark area shall be dimensions "E" (width) and "M" (length) as specified on Figures 4, 6, 7 and 8. Figures 2 and 9 shall have a document mark at the same location as Figure 4 (See 3.5.1).



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3.3.15.5 Timing Mark and/or Document Mark Area. The timing mark and/or document mark area shall be located in the microfilm area on the "E" (width) dimension specified on Figures 1, 4, 5, 6, 7 and 8. The timing mark (document reset mark) is used to reset the electronic counters so that the next mark will be counted on document count system. Timing marks are also repeated on Excess-Three-Binary Code systems to count the code. The code timing marks are not as long in length as the reset mark. All dimensions shall be inspected as specified in 4.6.9.

3.3.15.6 Bar Code. The bar code area shall be the first 16 of the 17 equally spaced positions "K" (width) and "G" (length) measured from the top of the "A" (0.520 inch dimension) as depicted on Figures 4 and 8. The seventeenth position is reserved for the frame number (See 3.3.15.3). The quantity of bars used to depict the frame location shall be selected from Figure 8.

3.3.16 Area Density.

3.3.16.1 Area Surrounding Opaque Code. The entire area surrounding the opaque code shall have a maximum density of 0.16.

3.3.16.2 Clear Area at Beginning and End of Film. All microfilm shall have 18 inches of clear film at the beginning and end of the roll as specified on Figure 1. The clear film shall be unexposed and processed.

3.3.16.3 Area Surrounding Clear Line Code. The density of the area surrounding the clear line code shall read not less than 0.90 and shall not be allowed to bleed (overlap) into the clear line code.

3.4 Documents.

3.4.1 Types. The documents that may be microfilmed are: specifications, drawings, sketches, lists, standards, pamphlets, reports, manuals, data sheets, abstracts, cards, letters, and any other form of data specified (See 6.2).

3.4.2 Reduction Ratio. All documents shall be exposed so that the microfilm frame contains the largest area of document coverage. Unless otherwise specified, all documents shall be microfilmed using the ratios depicted on Figure 10. In addition to the size of the document, the type point size also must be considered. (See 3.3.9, 3.3.9.1, 3.3.9.2). The lowest reductions that present American cameras can be set is 8X (16MM camera) 12X (35MM camera).



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3.4.3 Placement of Document Retrieval Codes. All documents shall be coded or uncoded as specified (See 6.2). The type of codes that shall be specified (See 6.2) are depicted in 3.3.11.1 through 3.3.11.9.1.

3.4.4 Image Dimensions Within Microfilm Frame. The image dimensions were selected, after careful study of present-day practices, to allow for a "B" size drawing to fit within the microfilm frame dimensions "A" and "B" allowing a 7.9% margin in the length dimension and a 11% margin in the width. The allowable original (hard copy) document sizes were generated by multiplying the reduction by the length and width of the permissible document image dimensions respectively. For example:

> 24:1 Reduction (Figures 4, 6 and 8)
> Document Width 24" x 0.464" ("C" dimension equals 11.136)
> Document Length 24" x 0.709" ("D" dimension equals 17.016)
> Therefore, for 24:1 reduction, the largest document that can be microfilmed and stay within the limits is 11.136 x 17.016 inches. (See Figure 10 for conversions of Figures 5 and 7.)

3.5 Copy Board.

3.5.1 Copy Board Layout Dimensions. The dimensions "A" and "B" on Figures 4, 6, 8 and dimensions "A-1" and "B-1" on Figure 5 and dimensions "A-2" and "B-2" on Figure 7 represent the frame area of the microfilm. The bottom of the "F" dimension on Figures 4, 6 and 8 and the "F-1" dimension on Figure 5 and the "F-2" dimension on Figure 7 represents the line where the top of the document to be microfilmed is placed. The document butts to the right edge of the microfilm frame. Dimensions "E" and "M" represent the document mark on Figures 4, 6, 8 and 9. Dimensions "E-2" and "M-2" represent the document mark on Figure 7. The proper color for the documents mark shall be color chip #37038 (black) or color chip #37875 (white) FED-STD-595, as specified (See 6.2). The document mark is centered under the frame area. The "G" and "K" dimensions represent the frame



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number area. The "G" and "K" dimensions also represent width and length of the code bar. All dimensions are determined by the document size as specified on 3.4.4. The copy board shall be the color chip #37875 (White) or color chip #37038 (Black) FED-STD-595.

3.5.2 Bar Code. Figure 8 may be used for making bar code. The document is placed on the copy board in the same manner outlined for 3.5.1. The quantity of bars required is depicted on Figure 8. The code size is cut from metal or paper using color chip #37038 (Black) or color chip #37886 (White) FED-STD-595. Place the frame number into the assigned area in accordance with 3.3.15.3 and 3.3.7. The bar(s) shall remain in the same position for twenty documents (only change frame number) unless otherwise specified (See 6.2).

3.5.3 Document Count Mark. Figures 4 and 7 may be used for making document count mark (image count, page count, document reset mark) film. The same procedures depicted in 3.5.1 may be used. In making document count film, every document count mark shall start with the first exposure; this will also include the frame number as specified on Figure 1 and 3.3.7.

3.5.4 Quantity of Documents Per Frame. Unless otherwise specified (See 6.2), the quantity of sheets or pages that may be included in a single microfilm frame is governed by the size of the document, camera aperture, width of microfilm, type point size and reduction ratio.

3.5.5 Comic or Cine Mode. As specified (See 6.2), the comic or cine mode method shall be permitted.

3.5.6 Document Title. The document title (title page) shall be placed on the frame as specified (See 6.2).

3.6 Container.

3.6.1 Type. The type of container required for distribution film shall be as specified (See 6.2).

3.6.2 Used. The government may furnish used microfilm containers as Government Furnished Material (GFM). Where GFM containers are furnished, the contractor shall inspect same for wear and tear. If the GFM containers are unserviceable, they shall be rejected.



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3.6.3 Containers Pre-Conditioning. All containers, new and used, shall be waxed FED-SPEC-P-W-120. The wax shall be buffed with a (lint free) cloth so that all visible traces of wax are no longer present. An alternate method of waxing would be to wax the film edge with the film wound tight on the core. Either method is acceptable, provided that the film loaded in the container can be moved freely without a bind. Waxing may not be required when specified in 6.2.

3.6.4 Rewind High Speed. After the container is loaded with film, it shall be placed in a reader or other simulating equipment. The entire roll of film shall be run through the reader. The reader shall then rewind the film on the container at high speed. This will also assure that the film is wound tight on the core (spool), and test the splicing of the leader and trailer.

3.6.5 Copy Film Container. Unless otherwise specified (See 6.2), all copy film (1.2) shall be loaded into the applicable container specified 3.6.7 - 3.6.11.

3.6.6 Types of Containers. The types of containers used to ship processed microfilm shall be of the following types:

3.6.7 Type C1. Type C1 shall be an Eastman Kodak type 2A.

3.6.8 Type C2. Type C2 shall be a 3-M Company type.

3.6.9 Type C3. Type C3 shall be a Stromberg-Carlson type (Datagraphic).

3.6.10 Type C4. Type C4 shall be a Bell & Howell type (Micro-Data).

3.6.11 Type C5. Type C5 shall be other as specified (See 6.2).

3.7 Reel. Reel in accordance with ANSI PH5.6-1966.

3.8 Targets.

3.8.1 Resolution. Figure 2 depicts the method for making a resolution target. The reading of the resolution for this specification shall be 24:1 reduction only. This one reduction is used



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due to the small size of the 16MM frame. The target is made from the National Bureau of Standard Microcopy Resolution Test Chart 1010a. The NBS target is cut and arranged on Figure 2 within the dimensions allowed.

3.8.2. Density. Figure 9 specifies the method of making a density target. This target shall be the same dimensions overall as specified in Figure 2. The 6% reflectance target shall be accurate within $\pm$ 0.3%. The density target shall be microfilmed at 24:1 reduction ratio and exposed to produce a density of 1.0 to 1.2 on the 50% reflectance target.

3.8.3 Security Marking. Unless otherwise specified (See 6.2), title targets indicating Security Classification shall be located on rolls as specified on 3.3.7. In addition to security markings such targets shall include a warning note as follows: This material contains information affecting the national defense of the United States within the meaning of espionage laws, Title 18, USC, Section 793 and 794.

3.9 Supporting Documentation. When specified (See 6.2), a hard copy index shall be furnished in the format specified by the procuring agency (See 3.3.14).

3.10 Security. Microfilm of classified documents shall bear the same classification as the original documents. Shipment of the microfilm must therefore be performed according to the security regulations applicable to the contract (See 3.8.3).

3.11 Reduction Bar. The resolution target Figure 2 and the density target Figure 9 are required to be microfilmed in sequence as specified in 3.3.7. Both targets shall be microfilmed at 24:1 reduction ratio and the reduction bar specified on Figure 9 shall be measured to determine if the 24:1 reduction ratio has been met.

## 4. QUALITY ASSURANCE

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.



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4.1.1 Certification of Tests. Certification of all tests performed in accordance with this section shall be submitted to the procuring activity on a contractor's form similar to Figure 11 and in accordance with contractual requirements.

4.2 Quality Conformance Inspection. The examinations and tests comprising quality conformance inspections are classified as specified in Table I. The rolls of processed microfilm shall be sampled and inspected in accordance with MIL-STD-105 and applicable subparagraphs hereto. Unless otherwise specified by the procuring activity (See 6.2), normal inspection shall be used at the start of the contract. Each type and class of microfilm shall be inspected and accepted or rejected separately. For the purpose of background density tests and visual inspection, an inspection lot is the total number of frames on the rolls of microfilm submitted for acceptance at one time. For purpose of resolution test, it is the number of rolls submitted for acceptance at one time. For the purpose of residual hypo test, an inspection lot is the number of rolls of microfilm processed and tested at one time.

TABLE I - EXAMINATIONS AND TESTS

| Tests | Requirement | | Test Method |
|---|---|---|---|
| Inspection | Req. Par | Test Par. | AQL or Sampling Plan |
| Visual Examination | 3.3 | 4.6.6 | Major .65%<br>Minor 2.5%<br>Control 4.0% |
| Reduction Ratio | 3.4.2 | 4.6.3 | Major .65%<br>2 sample units per lot. |
| Resolution | 3.3.8 | 4.6.4 | do |
| Background Density | 3.3.13.1 | 4.6.1 | do |
| Residual hypo | 3.3.1 | 4.6.2 | do |
| Spacing and Centering | 3.5.1 | 4.6.5<br>4.6.9 | do |
| Reproducibility and Readability | 3.3.9 | 4.6.7 | do |



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4.3 Microfilm. The original microfilm shall not be used to perform frame by frame inspection unless otherwise specified by procuring activity. The original microfilm may be used to check the density and resolution targets and security markings only. The frame by frame inspection for original microfilm shall be accomplished by using the copy film (See 4.4).

4.4 Copy Film. The copy film prior to delivery to the procuring activity, shall be inspected frame by frame at a minimum of 6X diameters of enlargement to determine retakes necessitated due to faulty camera operation, improper exposures of images, codes security markings, and any other deficiencies. If any of the above errors are detected during the inspection, the original roll will be remicrofilmed and additional distribution film generated in accordance with Section 3 unless an additional distribution copy roll can be made from the original that can pass inspection, proving that the errors were in the making of the copy and not in the original microfilm. (Inspect 1 roll only.)

4.5 Inspection.

4.5.1 Density Test. The density test shall be performed on microfilm in accordance with ANSI PH2.19-1959. The combined accuracy and precision of densitometers shall be so maintained that measurements of diffuse density, Type VI B, defined in ANSI PH2.19-1959 are within the following tolerances: $\pm$.01 density units at density up to and including 1.00 and $\pm$.015 density units at density levels greater than 1.00. This shall be accomplished by determining the accuracy of densitometers at least once every month at density levels in the range of .05 to .30 and 1.0 to 1.20 using standard density patches. If one or more of the readings fail to pass the test, the frame shall be rejected.

4.5.2 Residual-Hypo Test. The silver-halide microfilm shall meet the conditions for residual-hypo content as set forth in FED-STD-125 when tested by the method described in ANSI PH4.8-1971. If one or more of the samples fail to pass the residual-hypo test, the lot shall be rejected.

4.5.3 Reduction Ratio Test. The resolution target (Figure 2) and the density target (Figure 9) shall be microfilmed as specified in 3.3.8. The reduction ratio test strip shall measure 0.500 inches, plus or minus 0.002 inches. If the above dimensions are not met, the roll of microfilm shall be rejected.



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4.5.4 Resolution. A test shall be made of the resolution test target Figure 2. The microscope used to measure the target shall have a minimum magnification ratio of 50X. The target, Figure 2, shall be measured on the five positions. (If the camera is out of focus, the number of lines observed in some patterns of the image of the resolution chart may be different from five. This is known as "spurious resolution" and is usually accompanied by failure to resolve at one or more lower spatial frequencies.) If the directions of the lines on any of the five targets are not distinguishable or if there is evidence of spurious resolution, the roll shall be rejected. If there is no evidence of spurious resolution, the pattern may be examined to see if the lines are resolved. The resolution test shall be made and the resolution should read as depicted in 3.3.8.

4.5.4.1 Astigmatism. Astigmatism in the eyes of the observer may cause two groups of parallel lines which are oriented with the direction of the lines perpendicular to one another to appear unequally resolved. If an observer considers the lines resolved in only one of the two groups of the specified pattern, he should rotate the image (or his viewing position) 90° about the optic axis of the microscope and view the image again. If he then finds the lines resolved in a group where they had appeared not to be resolved, this test pattern may be considered to have met the resolution requirement. When neither of the patterns appear resolved, a rotation of 45° may cause one group to be resolved. A further rotation of 90° may then permit resolution of the other group and, if so, the resolution requirement is met. If not, the requirement has not been met and the roll shall be rejected.

4.5.5 Spacing and Centering. Three consecutive frames shall be selected at random without regard to quality on the roll of microfilm. If any of these frames fail to meet any of the spacing and centering test requirements, the roll of microfilm shall be rejected and counted as one defective.

4.5.6 Visual. Frames of microfilm shall be inspected at any one magnification from 6X through 24X for the defects listed in Table II using a microscope, reader and/or applicable electronic circuit for retrieving:



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TABLE II

Classification of Visual Defects

| Classification | Defects |
|---|---|
| Major | Document number is not clear and distinct. |
| | Characters or symbols are filled in or too light to the extent they are illegible. |
| | Lines are discontinuous or too light to the extent they are illegible. |
| | Data is obscured, illegible, omitted or out of focus. |
| | Blisters, tears or processing stains are on microfilm. |
| | Scratches are on microfilm, touching or through the document and/or code area. |
| | Foreign material is in document and/or code area, obliterating, obscuring or defacing document image area. |
| | Scratches, streaks, bubbles, lap marks, lumps or foreign material is over document image and/or code area obliterating, obscuring or defacing applicable areas. |
| | Document and/or code is not in its respective area. |
| | Images cannot be retrieved by applicable code. |
| | Documents not in sequence. |



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TABLE II (continued)

| Classification | Defects |
|---|---|
| Major | Security marking not correct. |
| | Wrong type container. |
| | Reduction ratio not correct. |
| | Index not on film. |
| | Document types, reduction ratio, incorrect. |
| | Hard copy layout, code layout, incorrect. |
| | Container type, pre-conditioning, incorrect. |
| | Film not rewound at high-speed. |
| Minor | Foreign material is in document image or code sector area, not obliterating, obscuring or defacing image or code area but adversely affecting appearance of frame. |
| | Scratches, streaks, bubbles, lap marks, lumps or foreign material are in protective coating over document image or code sector area, but not obliterating, obscuring or defacing image or code area. |
| Control | Finger marks, oil, grease are on microfilm. |
| | Scratches are on microfilm, not touching or through the document image or code section area. |



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TABLE II (continued)

| Classification | Defects |
|---|---|
| Control | Foreign material is in frame but not in document image or code sector area. |
| | Dust spots on background of document image or code sector area. |
| | Scratches, streaks, bubbles, lap marks, lumps or foreign materials are in protective coating but not over document image or code sector area. |
| | NOTE: All paragraphs are when applicable. |

4.5.7 Reproducibility Test. Two frames other than target shall be selected at random on the Type 1 Class 2 microfilm sample roll of microfilm. Paper prints shall be made using any enlargement ratio in the range of 14X through 30X. The two prints shall be examined for legibility. If both prints are not legible, the sample roll of microfilm shall be rejected and Government acceptance of microfilm shall be stopped until corrective action satisfactory to the Government has been taken.

4.5.8 Inspection of Microfilm by a Contractor Activity. Prior to delivery to the procuring activity, microfilm shall be inspected frame by frame at a minimum of 8 diameters of enlargement for retakes necessitated by faulty camera operation, improper exposure and other faults. Inspection shall be performed using instruments which will not scratch or otherwise damage the microfilm.

4.5.9 Dimensions. All dimensions depicted in Section 3 and applicable figures shall be considered in all inspections.

4.5.10 Remicrofilming Documents in Rejected Rolls. Documents in rejected rolls shall be remicrofilmed in accordance with Section 3.



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4.5.10.1 Rejection of Microfilm by a Government Activity. Unless otherwise specified by the procuring activity, when individual frames are rejected, the entire roll shall be remicrofilmed.

4.5.11 Certificate. The certificate, figure 11, has been included in this specification in order that it may be copied by the contractor. As each shipment of microfilm is inspected and approved by the contractor, one of these certificates shall be filled out and mailed to the address indicated in the contract.

## 5. PREPARATION FOR DELIVERY

5.1 Interior Packaging. Each roll of original camera microfilm type 1 class 1 shall be packaged in a metal or plastic reel. The reel shall be placed in a reel size fitting plastic or metal box which shall be marked to identify the reel.

5.2 Exterior Shipping Container. The collective rolls of microfilm shall be packed in exterior shipping containers conforming to PPP-B-636. The microfilm shipment shall be addressed as specified by the procuring activity.

5.3 Marking of Shipments. Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129 and marking shall include the information "Microfilm of Engineering Data". Interior packages containing original camera microfilm (5.1) shall also be marked with the additional note "Avoid finger marking or mishandling of this original camera microfilm."

## 6. NOTES

6.1 Intended Use. Subject to rights in data acquired under the contract, the microfilm acquired in accorcance with this specification are intended to satisfy Department of Defense requirements for the microfilming of engineering documents and the preparation of reproduction microfilm.



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6.2 Ordering Data: Procurement documents shall specify the following:

a. Title: MICROFILMING OF ENGINEERING AND RELATED DOCUMENTS, 16MM REQUIREMENTS FOR
Date of Specification: _____ 19__.
Specification number: MIL-M-46849 .

b. Microfilm Classification (See 1.2.1) Original camera master microfilm shall be coded as follows:

| Type | Class | Subclass | (Place "X" in proper blocks.) |
|---|---|---|---|
| I | 1 | N | [ ] |
| I | 1 | P | [ ] |

Distribution copy film shall be coded as follows:

| Type | Class | Subclass | (Place "X" in proper blocks.) |
|---|---|---|---|
| I | 2 | N | [ ] |
| I | 2 | P | [ ] |
| I | 3 | N | [ ] |
| I | 3 | P | [ ] |
| I | 4 | N | [ ] |
| I | 4 | P | [ ] |
| II | 2 | N | [ ] |
| II | 2 | P | [ ] |
| II | 3 | N | [ ] |
| II | 3 | P | [ ] |
| II | 4 | N | [ ] |
| II | 4 | P | [ ] |
| III | 2 | N | [ ] |
| III | 2 | P | [ ] |
| III | 3 | N | [ ] |
| III | 3 | P | [ ] |
| III | 4 | N | [ ] |
| III | 4 | P | [ ] |

c. Processing (See 3.3.1) shall be in accordance with FED-STD-125 Yes [ ], Other [ ] Specify_____.

d. Splicing (See 3.3.6) within other parts of the roll (Except on leader and trailer) permitted Yes [ ], No [ ].



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e. Frame number area size and location (See 3.3.7.1 and 6.2 u.) shall be as specified in Figure 4 Yes [ ], Other [ ] Specify_____.

f. 16MM Microfilm reduction ratio (See 3.3.9.1) shall be in accordance with Figure 10 Yes [ ], Other [ ] Specify_____.

g. 35MM Microfilm reduction ratio (See 3.3.9.2) shall be in accordance with Figure 19 Yes [ ], Other [ ] Specify_____.

h. Microfilm retrieval code (See 3.3.11 - 3.3.11.9.1)
(Place "X" in proper block.)
Uncoded Code Style S1 [ ]
Uncoded (Short Length) Code Style S2 [ ]
Bar Coded, Code Style S3 [ ]
Document Count, Code Style S4 [ ]
Document Count, (Short Length) Code Style S5 [ ]
Binary-Excess-Three, Code Style S6 [ ]
Bar Coded (Short Length) Code Style S7 [ ]
Binary-Excess-Three (Short Length) Code Style S8 [ ]
Bar Code Between Microfilm Frames Code Style S9 [ ]
Microfilm Frame Length and Width as Specified Code Style S10 [ ]

i. Microfilm document sequence (See 3.3.11.6) shall be in ascending page number sequence Yes [ ], Other [ ] Specify_____.

j. Index (See 3.3.14) A microfilm index shall be microfilmed Yes [ ], No [ ]. The index shall be hard copy and not microfilmed Yes [ ], No [ ]. The index shall be hard copy and microfilmed Yes [ ], No [ ]. The format is attached as_____.

k. The frame number (See 3.3.15.3) shall be dimensions "K" (width) and "G" (length) Yes [ ], No [ ], Other specify_____.



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l. The document mark (See 3.3.15.4) shall be dimensions "E" (width) and "M" (length) as specified on Figures 4, 6, 7, 8, 2 and 9 Yes [ ], Other [ ] Specify_____.

m. Document reduction ratio (See 3.4.2) shall be the same as "f" above.

n. Document retrieval codes (See 3.4.3) shall be the same as "h" above.

o. Hard copy layout dimensions (See 3.5.1) shall have color chip #37038 (Black) [ ] or color chip #37875 (White) [ ] or Other [ ] Specify_____.

p. Quantity of documents per frame (See 3.5.4) shall be: Specify_____documents.

q. Copy film container (See 3.6.5) shall be as follows:

Type C1 [ ]
C2 [ ]
C3 [ ]
C4 [ ]
C5 [ ]
None [ ]
Reel [ ]
Other [ ] Specify_____

r. Security marking (See 3.8.3) shall be as specified in Yes [ ], Other [ ] Specify_____.

s. Supporting documentation (hard copy index) (See 3.9) shall be as specified in "j" above.

t. Original (first generation) camera microfilm shall be mounted as follows:

container

Type C1 [ ]
C2 [ ]
C3 [ ]
C4 [ ]
C5 [ ]
Reel [ ]



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u. Microfilm Options (See 3.3.11.9 - 3.3.11.9.1)

Option 1. [ ] Frame number located over document mark. (See Figure 12 )

2. [ ] Frame number located to the left of and between microfilm frames (right reading). (See Figure 15 )

3. [ ] Frame number located to the left of and between microfilm frames, vertical reading. (See Figure 16)

4. [ ] Bar Code located to the left of and between microfilm frames. (See Figure 17)

5. [ ] Length of frame "A" and width "B" depicted by Figure 18 shall be
A = _____.
B = _____.

6. [ ] Bar Code shall advance every (Specify) _____ documents. (See 3.5.2)

v. The following microfilm modes comic or cine shall be used (See 3.5.4):
Comic [ ]
Cine [ ]

w. Document title shall be as follows:
_____
_____ (See 3.5.6).

x. Quantity of used containers that will be Government Furnished Material (GFM) shall be specified _____ (See 3.6.2).



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6.3 Definitions. All definitions applicable to words or phrases used in this specification are contained in Section 6 of MIL-M-38761.



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Preparing Activity

Army: MI

Project Number

EDMS-0027

Custodians:

Army: MI
Navy: AS
Air Force: 26
DSA: DH

Reviewer:

Army: MU, AV, WC, SM, GL, SL, EL, TM
Navy: OS, EC, YD, SH, SA, MC
Air Force: 26
US Coast Guard: FS
DSA: SCT
NSA: N23
DASA: DS

User:

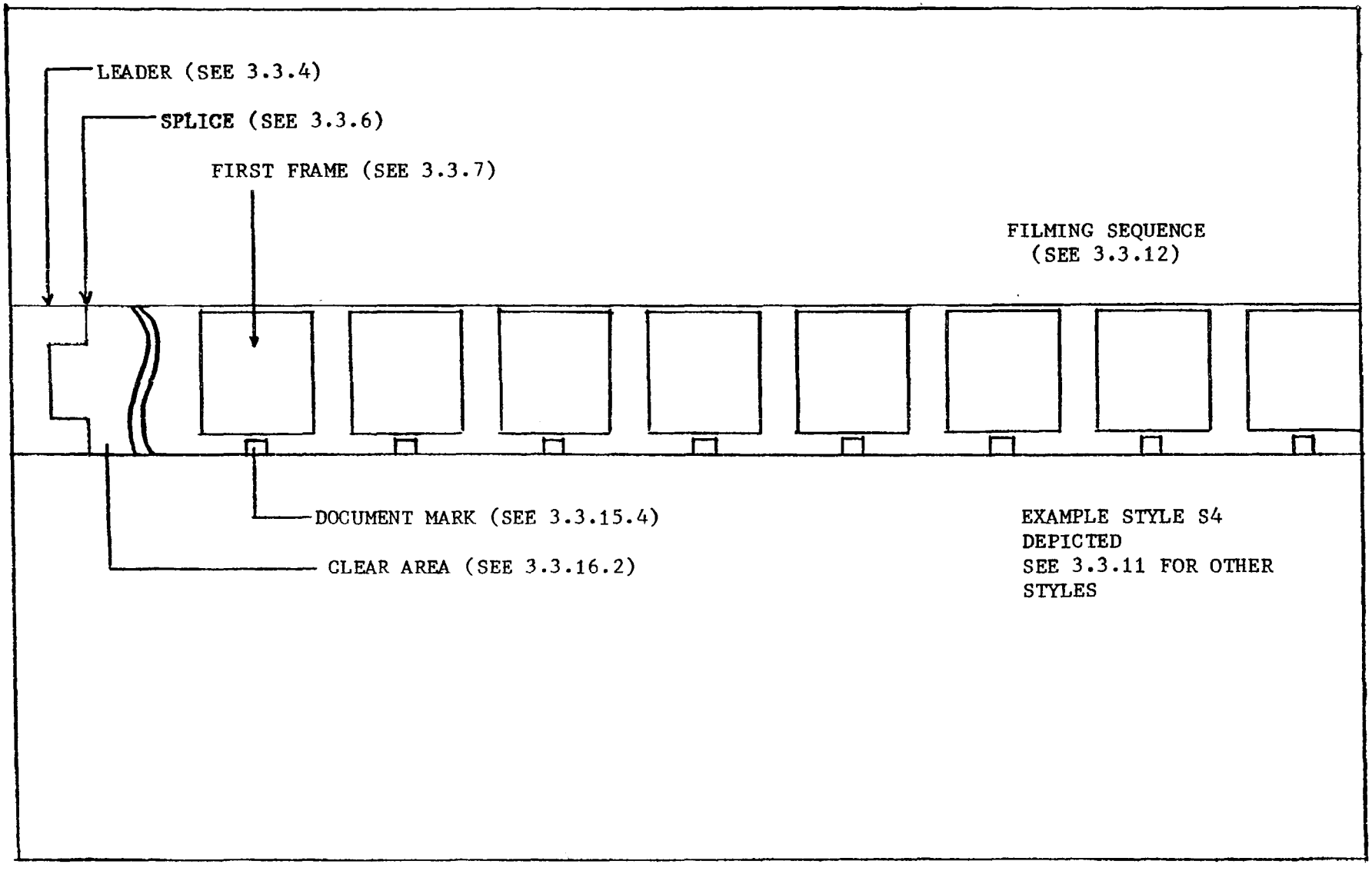
Army: TA

Industry

AIA
AOA
EIA
ANSI
NMA



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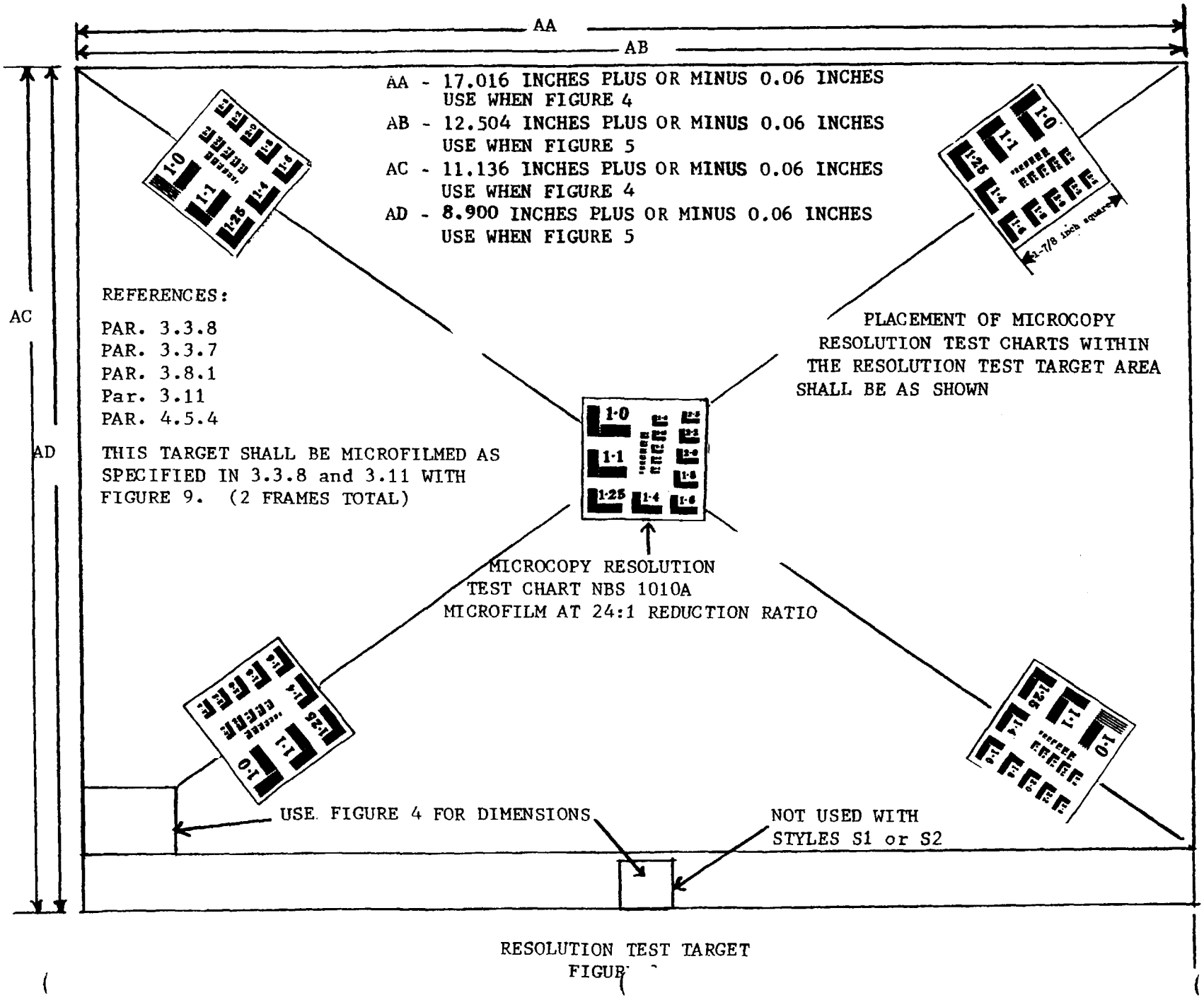

MICROFILM ARRANGEMENT

FIGURE 1





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RESOLUTION TEST TARGET
FIGURE



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:04Z
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| TYPE POINT SIZE | HEIGHT OF FONT DIMENSIONS INCHES (+.005) |
|---|---|
| 4 | L - .035<br>S - .025 |
| 6 | L - .055<br>S - .030 |
| 7 | L - .060<br>S - .040 |
| 8 | L - .070<br>S - .045 |
| 9 | L - .080<br>S - .050 |
| 10 | L - .090<br>S - .055 |
| 11 | L - .095<br>S - .060 |
| 12 | L - .105<br>S - .065 |
| 14 | L - .125<br>S - .075 |

L - Equals upper case letters and numerics plus b,d,f,g,h,j, k,l,p,q,t, and y.

S - Equals the following lower case letters: a,c,e,i,m,n,o, r,s,u,v,w,x, and z.

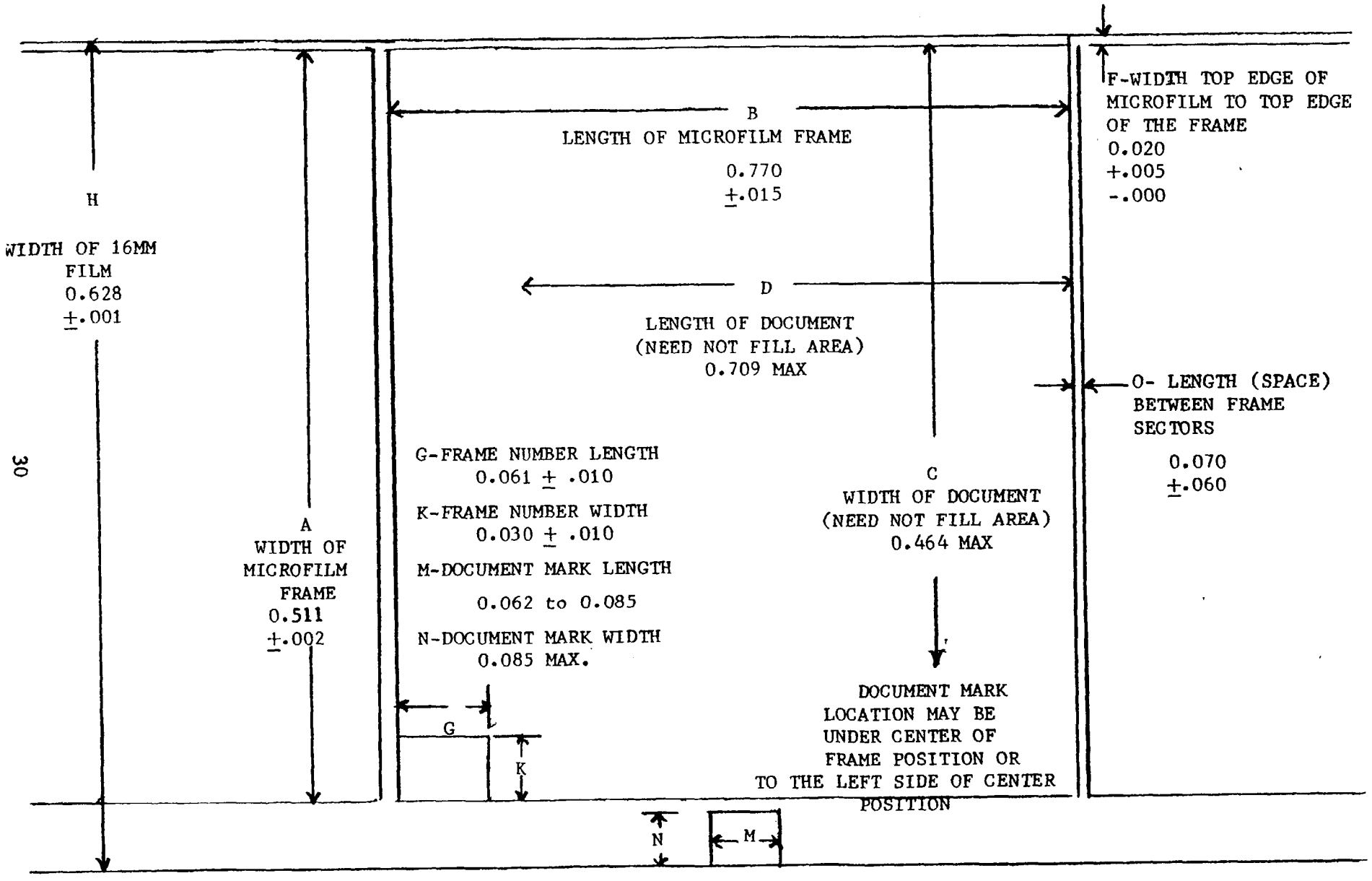
(See 3.3.12)

TYPE POINT SIZES

FIGURE 3



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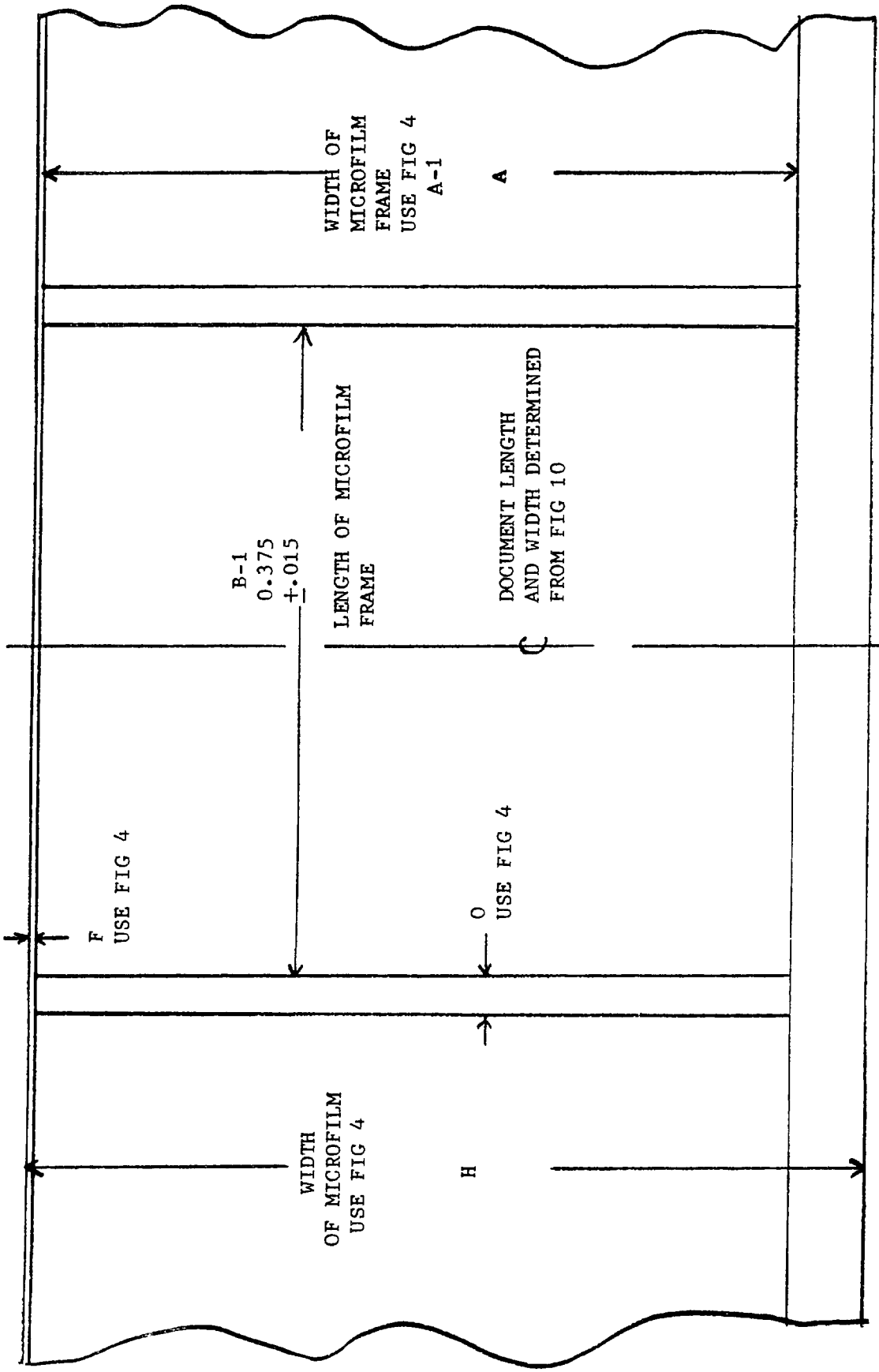



UNCODED. CODE STYLE S1 AND DOCUMENT COUNT. CODE STYLE S4

FIGURE 4.

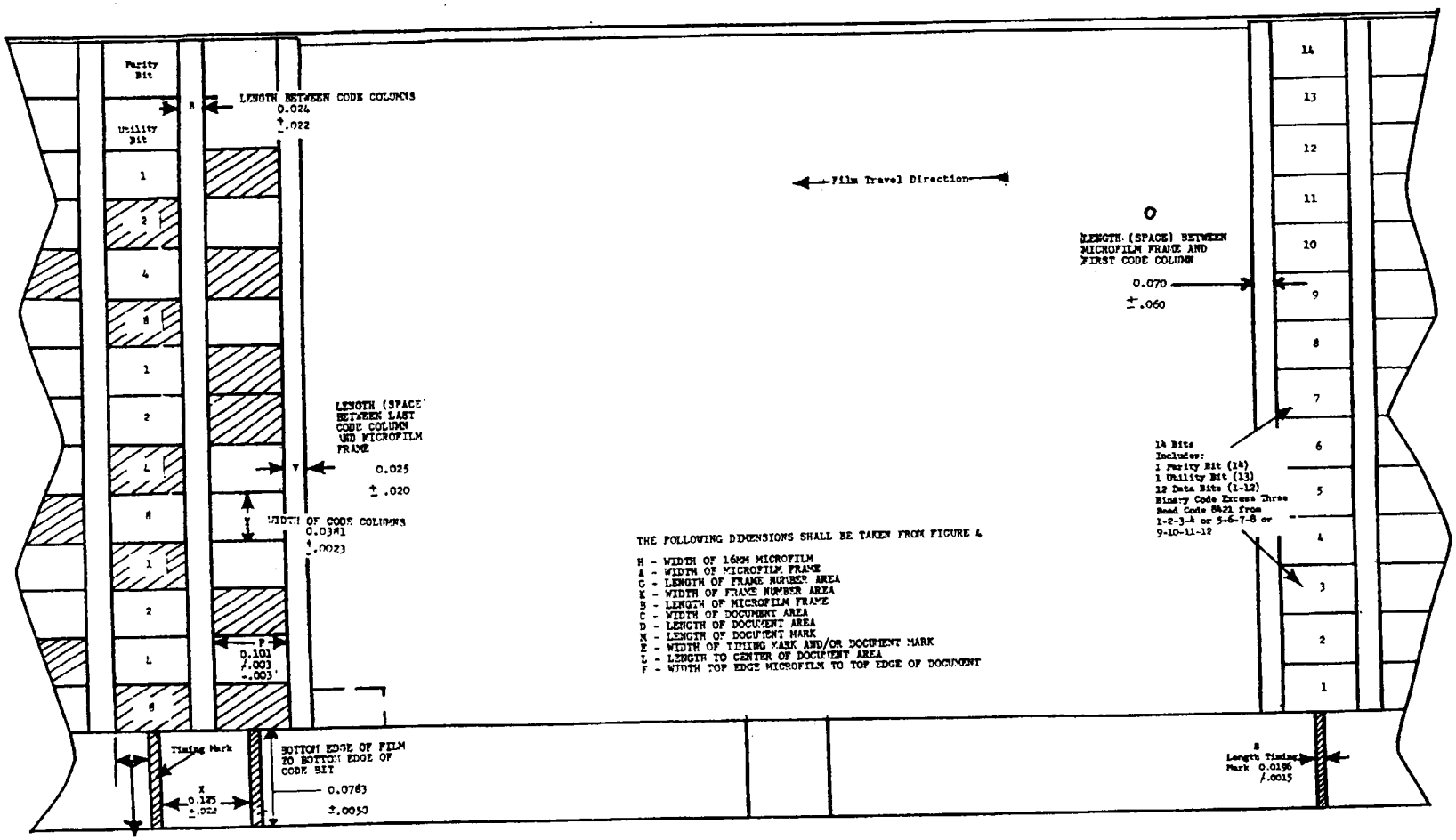


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UNCODED (SHORT LENGTH) 16MM MICROFILM
STYLE S2
FIGURE 5

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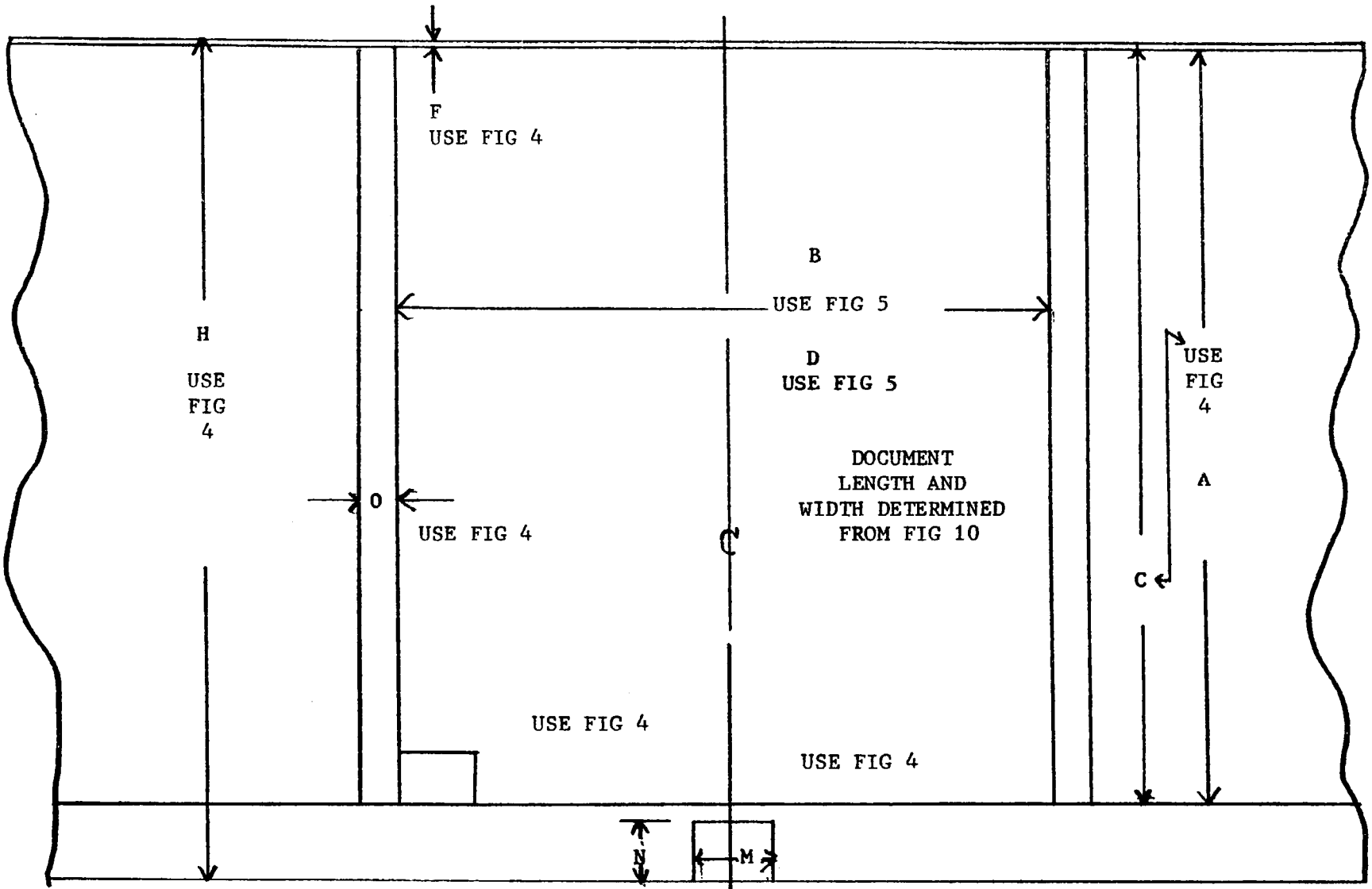



BINARY-EXCESS-THREE CODE, SYTLE S6, MICROFILM

FIGURE 6



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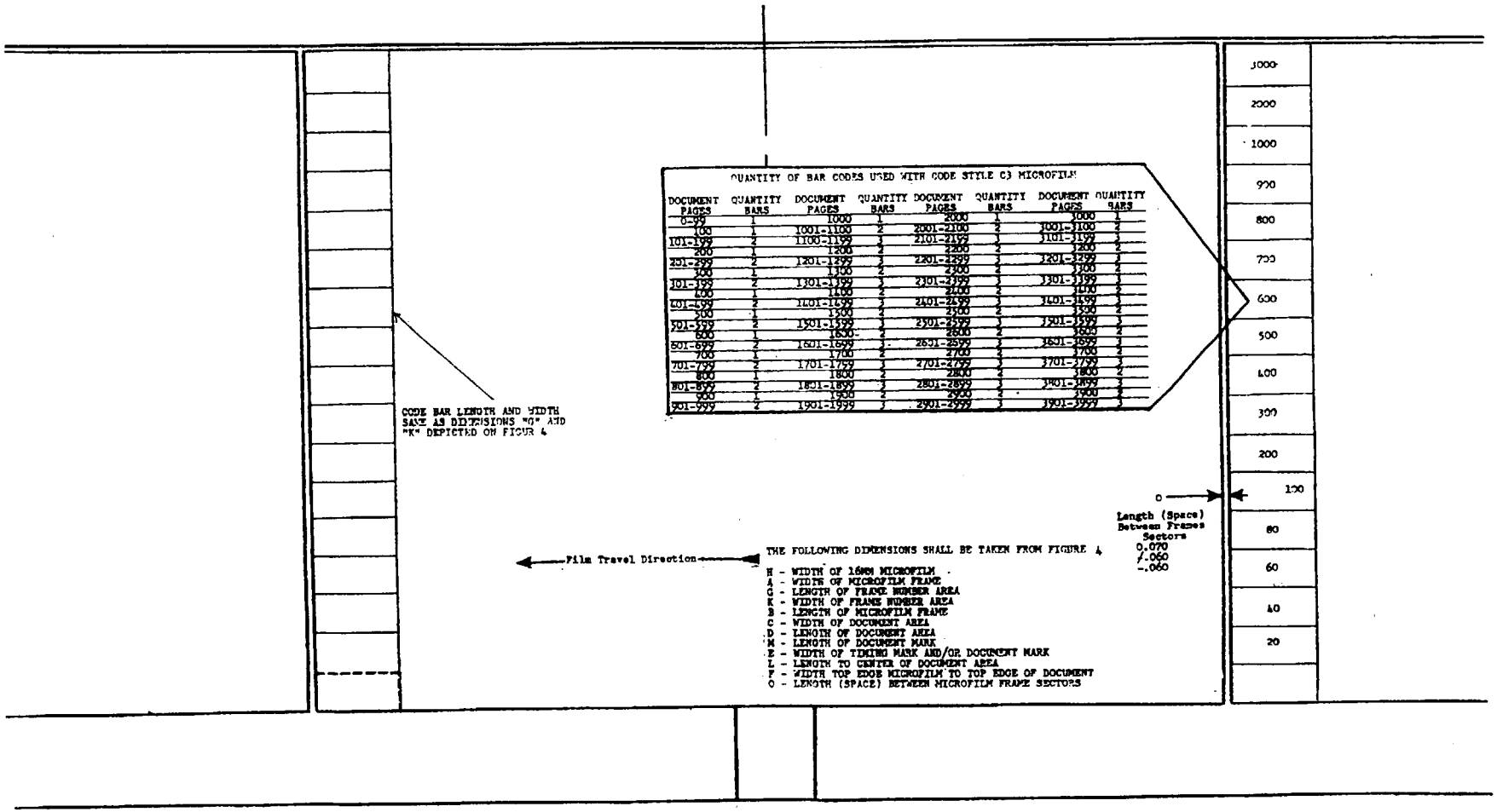



DOCUMENT COUNT (SHORT LENGTH) 16MM MICROFILM

STYLE S5
FIGURE 7



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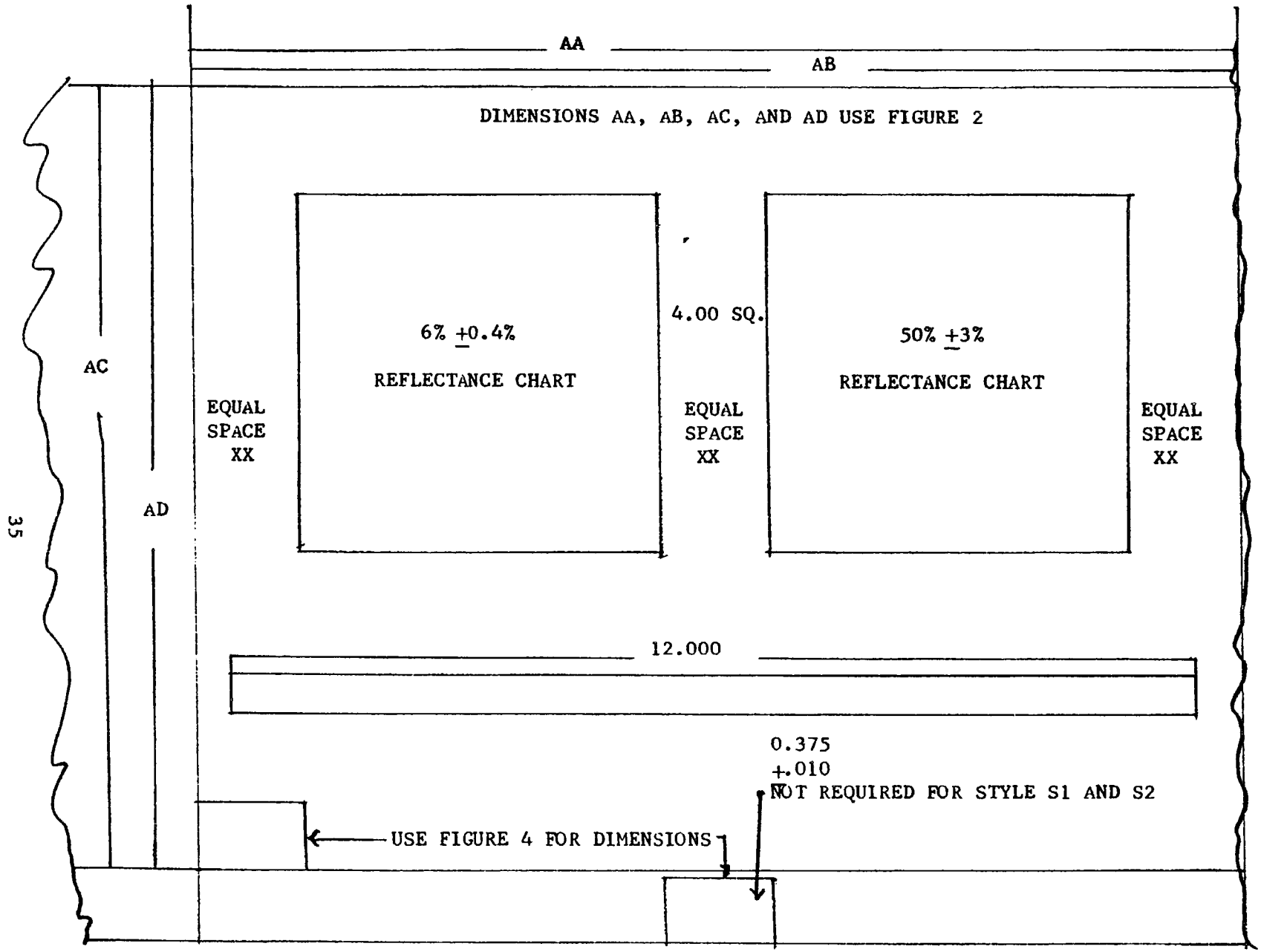



BAR CODE, CODE STYLE S3 MICROFILM

FIGURE 8



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DENSITY TEST TARGET

FIGURE 9





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## 16MM MICROFILMING REDUCTION RATIOS

The reduction ratio for microfilming of documents shall be determined from the document sizes listed below. All ratios larger than 24:1 are limited by the size of the copy board and the camera ratio.

| Camera Reduction Ratio | Document Size Fig 4,6,8. Width (0.464) | Length (0.709) | Document Size Fig 5 & 7 Width (0.520) | Length (0.375) |
|---|---|---|---|---|
| 8 ..... | 3.712 | 5.672 | 4.160 | 3.000 |
| 9 ..... | 4.176 | 6.381 | 4.680 | 3.465 |
| 10 ..... | 4.640 | 7.090 | 5.200 | 3.750 |
| 11 ..... | 5.140 | 7.799 | 5.720 | 4.125 |
| 12 ..... | 5.468 | 8.508 | 6.240 | 4.500 |
| 13 ..... | 6.032 | 9.217 | 6.760 | 4.875 |
| 14 ..... | 6.496 | 9.926 | 7.280 | 5.250 |
| 15 ..... | 6.960 | 10.635 | 7.800 | 5.625 |
| 16 ..... | 7.424 | 11.344 | 8.320 | 6.000 |
| 17 ..... | 7.888 | 12.053 | 8.840 | 6.375 |
| 18 ..... | 8.352 | 12.764 | 9.360 | 6.750 |
| 19 ..... | 8.816 | 13.471 | 9.880 | 7.125 |
| 20 ..... | 9.280 | 14.180 | 10.400 | 7.500 |
| 21 ..... | 9.744 | 14.886 | 10.820 | 7.875 |
| 22 ..... | 10.208 | 15.598 | 11.440 | 8.250 |
| 23 ..... | 10.672 | 16.307 | 11.960 | 8.525 |
| 24 ..... | 11.136 | 17.016 | 12.504 | 8.900 |

Minimum type size shall be: 8:1 - 20:1 4pts.
20:1 - 24:1 5pts.

FIGURE 10



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CONTRACTORS CERTIFICATE

I hereby certify that the roll(s) of microfilm included in this shipment have been inspected and approved as meeting the quality assurance provisions checked below and all other requirements of Specification MIL-M-46849

CONTRACTOR_____ CONTRACT NUMBER_____

MICROFILM ROLL(S) NUMBER_____THRU_____

GOVERNMENT AGENCY_____

DATE OF SUBMITTAL_____

RESOLUTION TEST.....____

RESIDUAL HYPO TEST.....____

REDUCTION RATIO.....____

BACKGROUND DENSITY TEST.....____

FRAME BY FRAME CHECK.....____

REJECTED FRAMES WILL BE
REFILMED AND SUBMITTED
LATER.....____

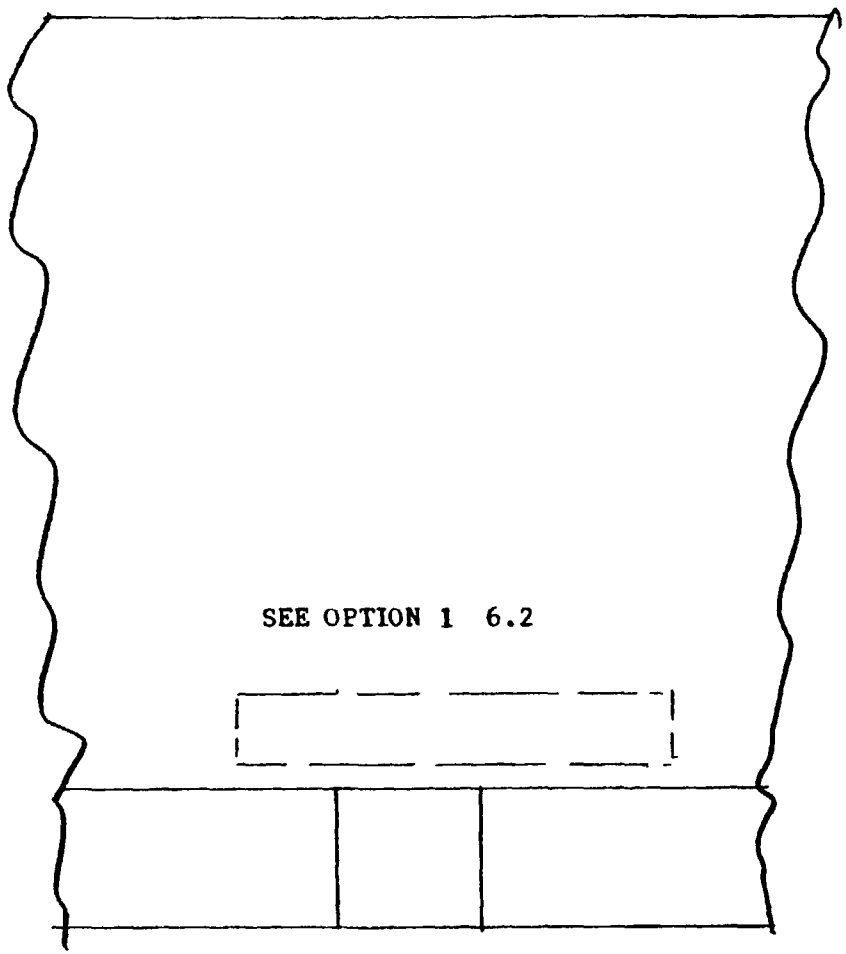
REJECTED FRAMES REFILMED
AND SUBMITTED HEREWITH.....____

FORM OF CONTRACTORS CERTIFICATE

FIGURE 11



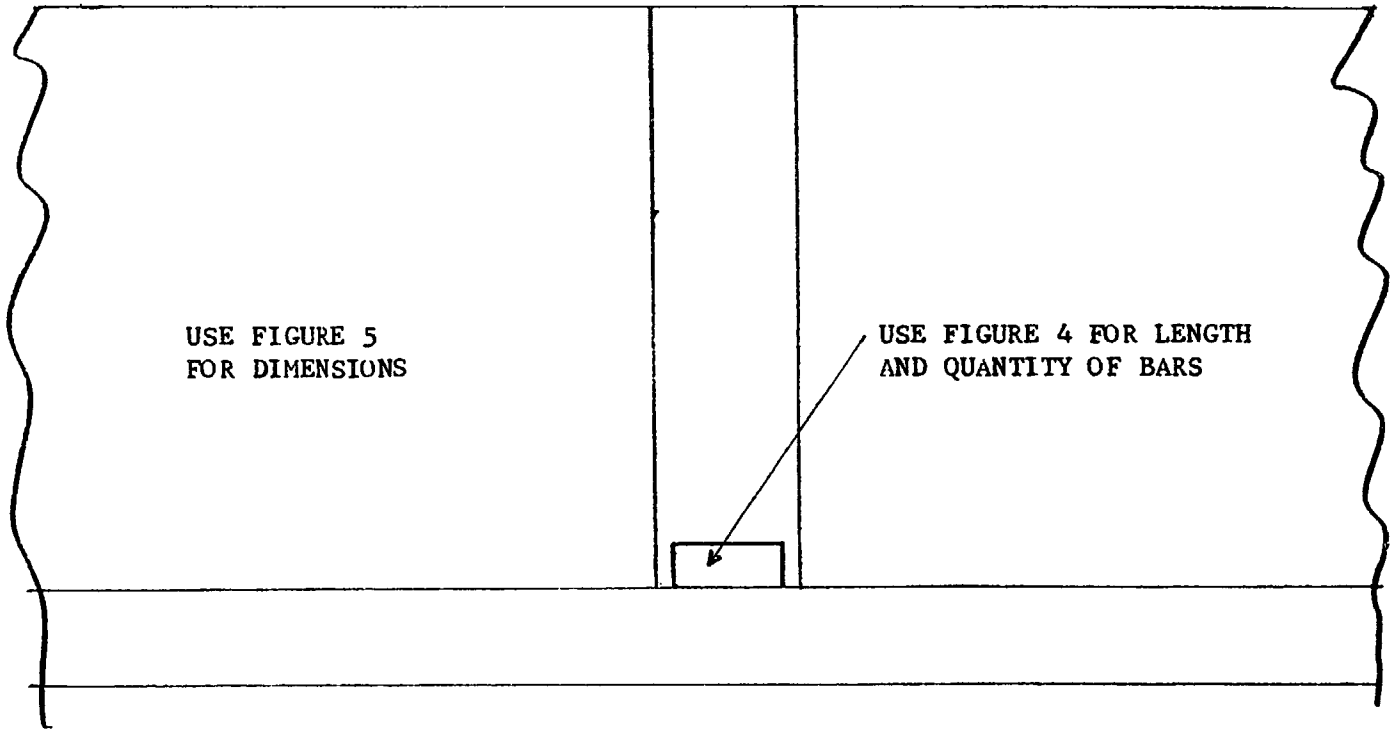
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OPTION 1

FIGURE 12

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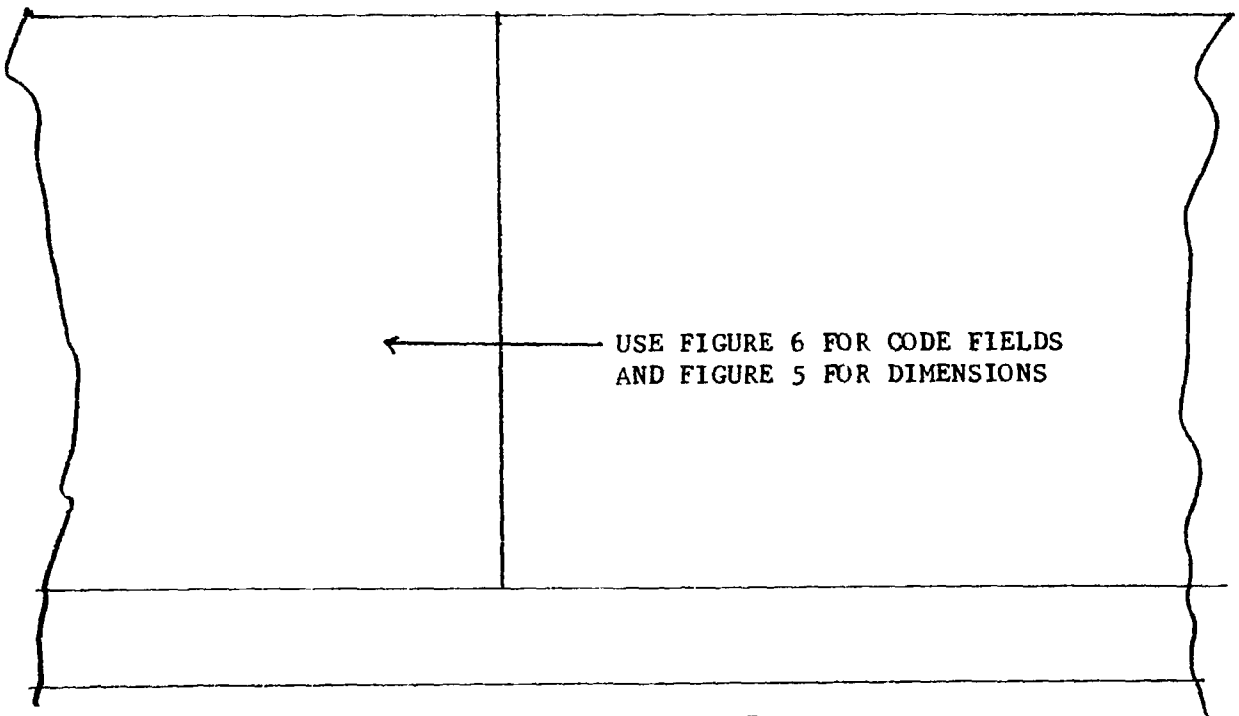

BAR CODE SHORT LENGTH

FIGURE 13





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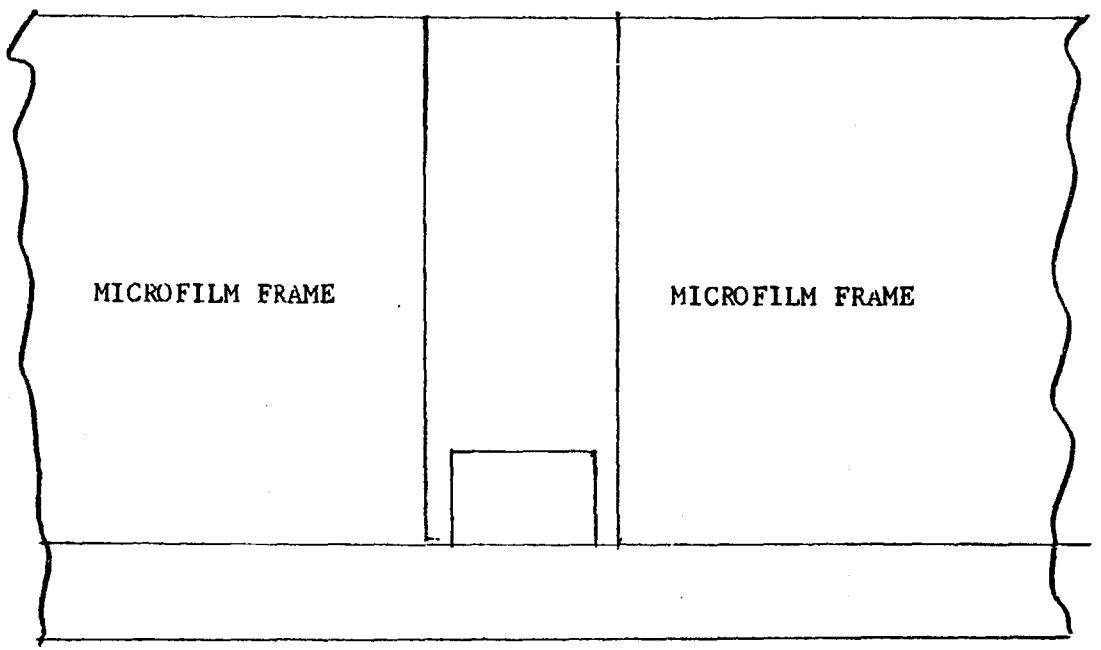



BINARY EXCESS THREE CODE

FIGURE 14



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OPTION 2

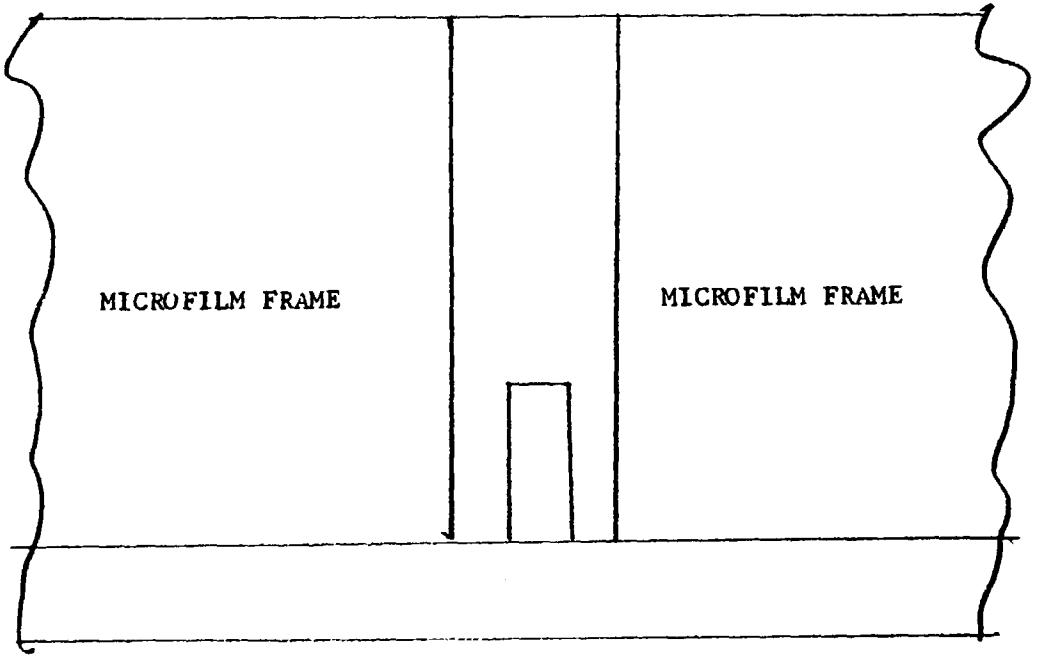
FRAME NUMBER LOCATED TO THE LEFT
OF MICROFILM FRAME (RIGHT READING)

FIGURE 15





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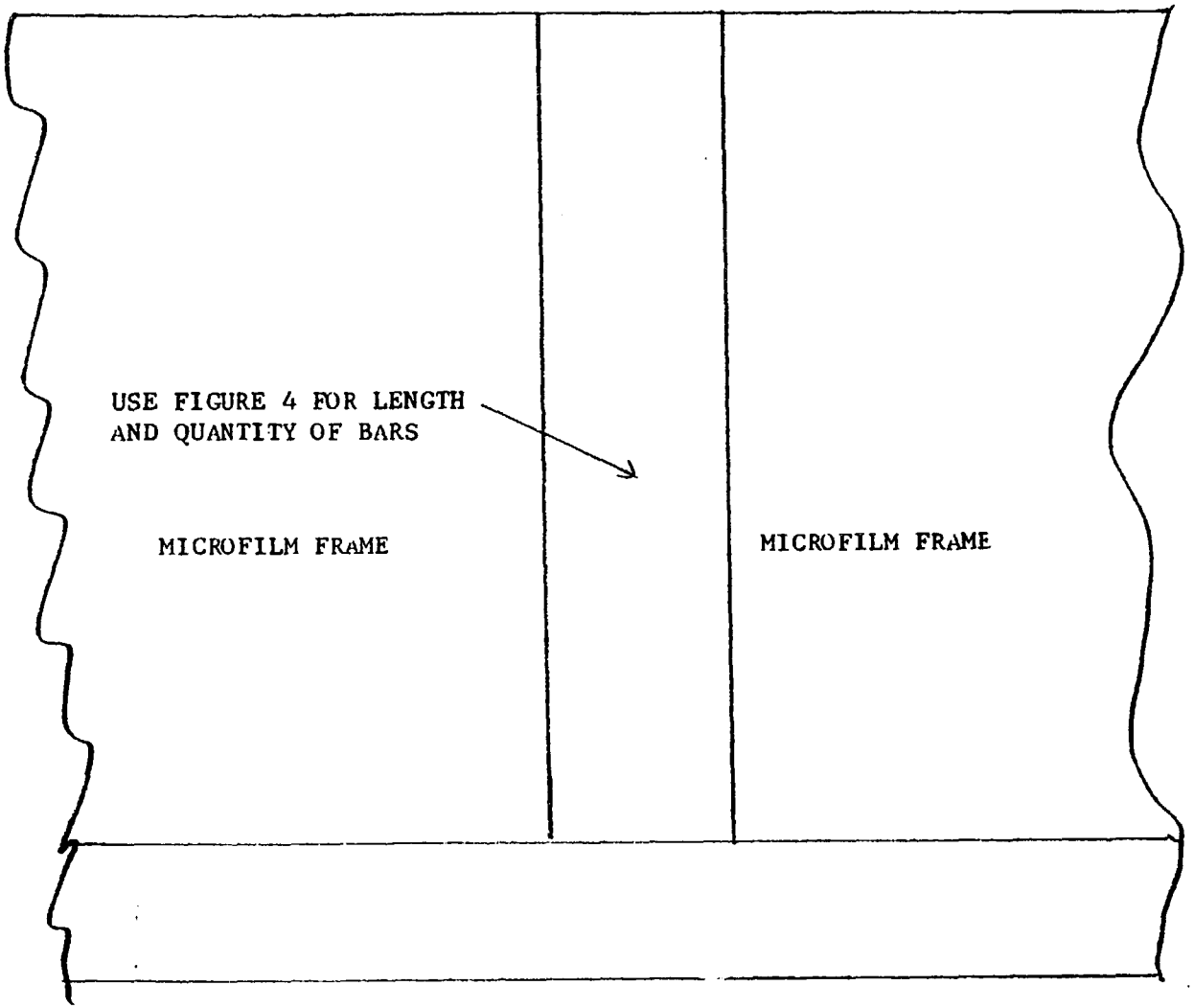



OPTION 3

FRAME NUMBER LOCATED TO THE LEFT OF THE MICROFILM FRAME (VERTICAL READING)

FIGURE 16

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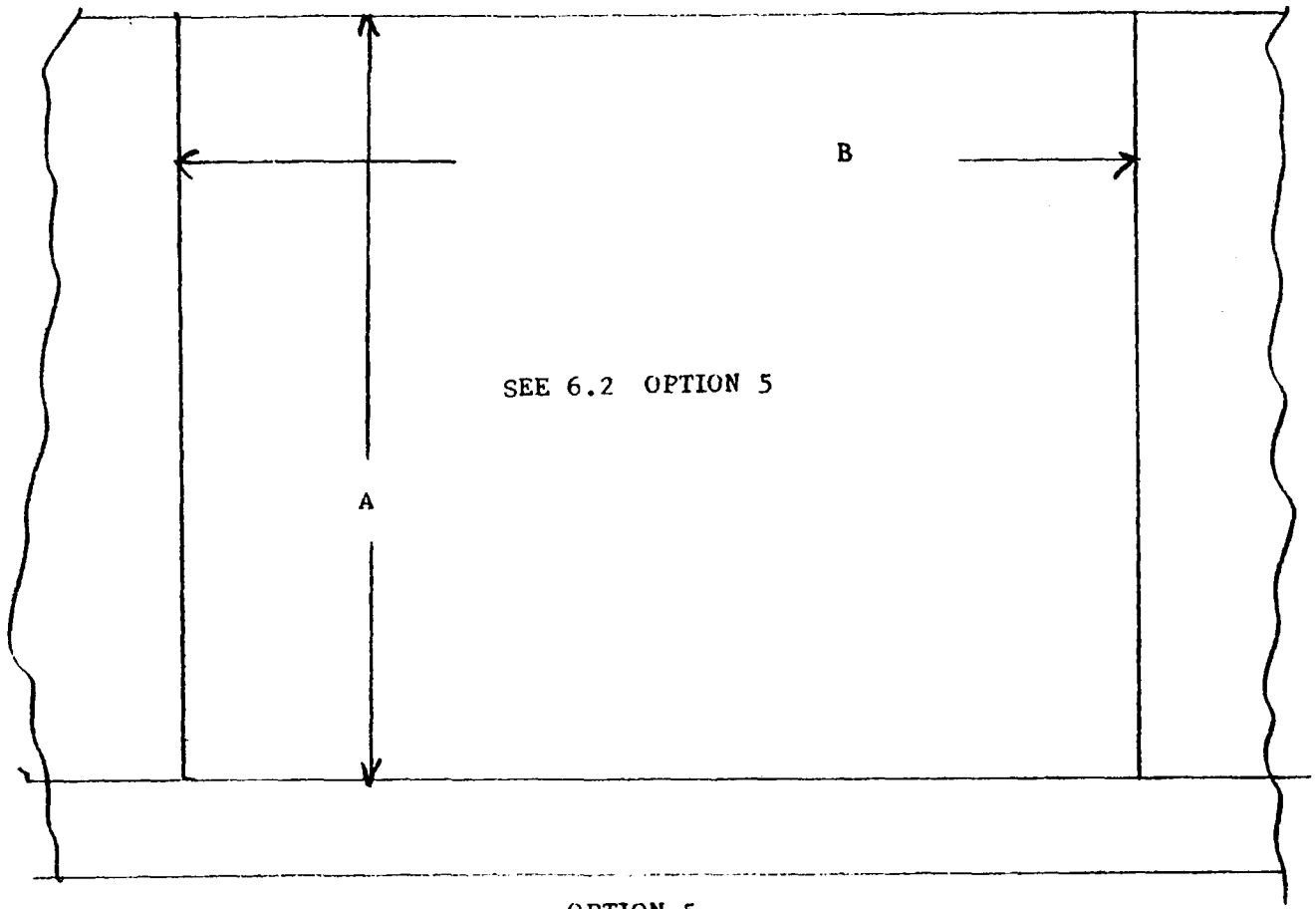

OPTION 4

BAR CODE BETWEEN MICROFILM FRAMES

FIGURE 17



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OPTION 5
NONE STANDARD LENGTH AND WIDTH

FIGURE 18

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## 35MM MICROFILMING REDUCTION RATIOS

The reduction ratio for the 35MM microfilming of documents when used to convert to 16MM must take into consideration the 0.464 width and the 0.709 length for Figures 4, 6 and 8. If the 35MM microfilm is used to generate 16MM microfilm for Figures 5 and 7 then the consideration of the 0.520 width and the 0.375 length must be considered. The lense ratio in the optical converter must also be considered in order to make the conversion. Minimum type point sizes shall be 8:1 6pts, 9:1-10:1 7pts, 11:1 - 12:1 8pts, 13:1 - 14:1 9pts, 15:1 - 16:1 10pts and 18:1 - 19:1 11pts.

FIGURE 19



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-M-46849 | MICROFILMING OF ENGINEERING AND RELATED DOCUMENTS, 16MM REQUIREMENTS FOR |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR<br>☐ USER<br>☐ MANUFACTURER<br>☐ OTHER *(Specify)*: _____ |
| b. ADDRESS *(Street, City, State, ZIP Code)* | |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

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